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Learning Tonotactic Patterns with Autosegmental Representations

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Abstract
This paper presents and evaluates an algorithm which learns tonotactic patterns over autosegmental representations (ARs), facilitated in part by showing how adjacency matrix representations can represent ARs. The Bottom-Up Factor Inference Algorithm over ARs (BUFIA-AR) takes AR graphs as input, and outputs a set of inviolable subgraphs as constraints. The input AR graphs themselves are generated automatically from linear transcriptions. A corpus of 621 non-derived native Hausa words was used to test BUFIA-AR, which returned 11 constraints. The results fall into three categories: constraints that align with previous generalisations, constraints that provide more specific generalisations than previously posited and which better account for the data, and constraints that identify novel patterns. These results demonstrate the feasibility and efficiency of ARs as a computational representation for phonotactic learning.

Contents

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1. Introduction

This paper provides one method for learning tonotactic patterns over autosegmental representations (ARs). A key insight of phonological theory is to view forms in tonal languages as *autosegmental graphs*, rather than one-dimensional linear strings (Goldsmith, 1976; Coleman and Local, 1991; Jardine, 2017; Oakden, 2020). These representations have been argued to have demonstrated advantages for analysing tonal patterns (Hyman, 2009; Clements and Goldsmith, 2010; Odden, 2013). Computationally, grammars over ARs have been shown to express both local and long-distance constraints, while maintaining a relatively low level of computational complexity (Jardine, 2017; Chandlee and Jardine, 2019). However, despite the importance of ARs in tonal phonology and previous research on their formal properties (Kornai, 2007), there has been limited work on learning tonal grammars, and no phonotactic learning models to our knowledge have adopted ARs.

This paper focuses on learning tonotactic grammars. Following Jardine (2017), the tonotactic grammar consists of a set of language-specific inviolable constraints over ARs. The constraints themselves only specify marked AR structures. Logically, such grammars are simply the conjunction of negative literals (Rogers and Lambert, 2019). In this setting, the tonotactic learning problem boils down to identifying those constraints (the marked AR structures) which are consistent with the entire dataset.

The Bottom-Up Factor Inference Algorithm (BUFIA) is a learning algorithm that deterministically searches a space of possible constraints and is provably guaranteed to find the most general constraints consistent with its input data (Chandlee et al., 2019; Rawski, 2021). Crucially, it is couched in model theory (Rogers, 1998; Enderton, 2001) so that it can operate with any particular representational scheme.

This work presents BUFIA-AR, an instantiation of BUFIA that learns tonotactic constraints expressed with ARs. As will be explained, this implementation utilises a matrix-based representation for ARs, which is an adaptation of matrix-based representations of graphs (West et al., 2001; Cormen et al., 2022).

BUFIA-AR was tested on a curated corpus of Hausa (Awagana et al., 2009), a Chadic language in West Africa. The corpus consists of 621 mono-morphemic words, which altogether provides a collection of 64 distinct ARs. Using this corpus as input, we assess the performance of BUFIA-AR through a combination of qualitative and quantitative evaluations. Qualitatively, BUFIA-AR returns a grammar consisting of 11 AR constraints when syllable, tone, and mora counts are limited to three. These BUFIA-AR-learned constraints either align with, or are more specific than, those previously proposed in the literature. The algorithm additionally identifies novel patterns that have not been previously reported. Quantitatively, a cross-validation experiment shows that BUFIA-AR achieves a mean accuracy over 90% in both token- and type-based settings which outperforms the linguistic baseline grammar and a BUFIA learner using string representations (BUFIA-Str). Altogether, these results demonstrate the feasibility of using BUFIA-AR for tonotactic grammar inference. All data and programs used in this paper are accessible on GitHub.

In this way, this paper takes a first step towards learning tonal grammars expressed in terms of ARs. We acknowledge that tonal grammars are much more than just the tonotactic component. The grammars that BUFIA-AR returns have nothing to say about how underlying representations of tonal languages are learned, nor how these underlying representations are transformed into surface representations. We also acknowledge that some surface constraints have been argued to follow from the tonal processes themselves in some languages (see Section 2), which is again something the grammars returned by BUFIA-AR cannot express. Nonetheless, despite these limitations, BUFIA-AR still makes a contribution to our understanding of how tonal grammars expressed in terms of ARs can be learned. We expect that some of the insights made here will persist in future research which addresses these broader problems regarding the learning of tonal grammars.

This paper is structured as follows. Section 2 provides an overview of the tonotactic learning problem and current models. Section 3 introduces formal language theory, model theory, and graph theory relevant to ARs and BUFIA. Section 4 provides a detailed description of the BUFIA-AR implementation. Section 5 presents the Hausa case study and model evaluation. Section 6 discusses issues relating to constraint optimisation, constraint satisfaction, and constraint weighting. Section 7 concludes the paper and points to some future directions.

2. Learning Tonotactics

2.1. Tonal Mapping Problem

A central question in tonal phonology concerns how languages organise tones, syllables, and/or moras into well-formed structures. A closely related modelling question asks what kinds of well-formed tonal patterns are attested in a language, and how such patterns can be systematically characterised. To illustrate these issues, we begin with a frequently discussed example: the tonal mapping problem in two West African tonal languages, Hausa and Mende. Hausa is a Chadic language, while Mende is a Southwestern Mande language spoken in Sierra Leone (Leben, 1978). Both languages use High (H) and Low (L) tones as basic tonemes with the mora as the tone bearing unit (TBU). However, they show distinct patterns in how their tonal melodies are mapped to TBUs (Newman, 2002; Leben, 1978; Zoll, 2003; Jardine, 2017).

	Mende			Hausa	
contour inventory	H L mbu 'owl'	L H L ngewɔɔ 'God'	L H mbaa 'rice'	H L mai 'oil'	
tone mapping	H L f e l a l a 'junction'		L H l e l e m a 'mantis'	H L j ee m aa g ee 'bat'	L H j i m i n a 'ostrich'

Table 1: Contour inventories and tone mapping in Mende and Hausa.

As shown in Table 1, the two languages first differ in their monosyllabic contour inventory. Mende permits a wider range of monosyllabic contours, including light HL, heavy HL, and heavy LH, whereas Hausa allows only the heavy HL contour. Secondly, when a two-tone melody maps onto a trisyllabic word, Mende and Hausa have distinct surface forms: HL surfaces as H.L.L in Mende (*fé.là.là* 'junction') but as H.H.L in Hausa (*jée.máa.gèe* 'bat'); similarly, LH surfaces as L.L.H in Mende (*lè.lè.má* 'mantis') but as L.H.H in Hausa (*jì.mí.ná* 'ostrich')[1,2].

There are mainly three approaches to the tone mapping problem. The first identifies three key factors that determine the surface forms: *directionality* (whether tone spreads rightwards or leftwards), *morphological categories* (prefix or suffix), and *tone quality* (whether spreading is sensitive to H or L) (Hyman, 1987; Leben, 2017; Hyman, 2009).

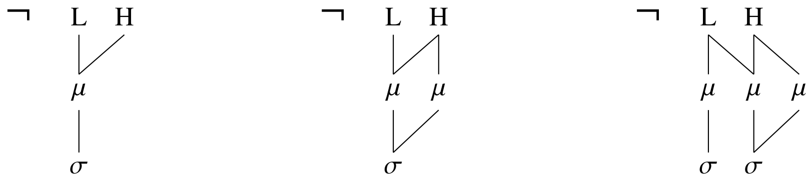
The second account is Optimal Tone Mapping (OTM; Zoll, 2003). Zoll argues that directionality-based accounts of phonological mapping can in some cases yield incorrect surface predictions. Under OTM, surface tonal patterns emerge from language-specific rankings between morphological faithfulness constraints and markedness constraints, Clash and Lapse, which are sensitive to tone quality.

The third approach, proposed by Jardine (2017), is to use language-specific, inviolable constraints, where the constraints are marked, forbidden ARs; that is, ARs to be avoided. For example, the No Short Rise constraint in Mende (Leben, 2017) that bans monomoraic LH can be represented as a banned AR as in Figure 1a.

Following Jardine (2017), if Figure 1a is forbidden in the language, any AR *containing* 1a, such as 1b and 1c, will also be forbidden. Formally, if Grammar G is the set of constraints in the language, such that $G = \{AR_1, AR_2, \ldots, AR_n\}$, a well-formed We represent factors as *conjunctive negative literals*: $\neg AR_1 \wedge \neg AR_2 \wedge \ldots \wedge \neg AR_n$ (Jardine, 2017; Rogers and Lambert, 2019). It follows that the tonotactic learning problem

[1] This paper follows the tonal transcription method commonly used by Africanists. For instance, "H.H.L" represents two H-toned syllables followed by a L-toned syllable. In addition, tones are always marked on the nuclear vowels. Long vowels are transcribed as double vowels.

[2] Dwyer (1978) offers a different analysis of Mende tones than Leben (1978)'s, and provides supplemental data in which L.H.H forms also occur (e.g. *ndàvúlá* 'sling'). Leben (1978) account for the different patterns by proposing two distinct mappings.



(a) $*\check{\mu}$ (b) $*\check{\mu}\acute{\mu}$ (c) $*\grave{\mu}\check{\mu}\acute{\mu}$

Figure 1: The forbidden structure in (a), $*\check{\mu}$ (corresponding to No Short Rise), and two ARs in (b–c) containing it.

is to identify a set G of banned ARs given a set of example surface forms. This paper follows the third approach of Jardine (2017).

One potential criticism of Jardine (2017), anticipated by the author and also applicable to the present work, is that it explains tonal patterns exclusively in terms of surface-true constraints. One consequence is that well-formed patterns that are not directly observable in surface forms, or at the lexical level, are not realised by the grammar. The grammar inferred at this level therefore reflects only the final outcomes of tonal mapping, rather than intermediate representations produced by tonal rules, which potentially misses important phonological generalisations. **[Do we want to mention ICGI paper here - maybe we should, why not]**

Hausa provides a clear illustration of this point. The absence of rising tones on a single syllable can be explained as follows. An underlying LH sequence may arise as an intermediate representation, either through synchronic phonological processes or as the result of historical change, but it is subsequently simplified to either H in non-final position or L elsewhere (Leben, 1971; Newman, 2002). Likewise, the absence of final L.L sequences on long vowels is attributed to Low Tone Raising, which changes underlying L.L to L.H when the second L occurs on a long vowel (Leben, 1971, 1996). **[I removed the SPE styled rules and explained in prose instead.]**

This example illustrates that unattested surface patterns do not necessarily reflect illicit structures in the grammar itself, but rather intermediate representations that are subsequently subject to some repair processes. While acknowledging the limitations of these grammars, we argue that learning such grammars is nonetheless valuable. They still provide meaningful insights into tonotactic patterns and serve as a foundation for further work for three reasons.

First, a grammar containing a set of forbidden ARs that is consistent with observable surface forms is not in conflict with a grammar for the tonal processes; rather, it captures their pronounceable outcomes. Given available access to a lexicon at a different level of abstractness, another grammar could likewise be learned, and these two grammars can be systematically compared. This comparison between different levels of abstractness itself yields non-trivial insights into tonal processes.

Second, considering that ARs have been long used in tonal phonology, yet absent from computational learning, we believe a more pressing priority is establishing representational adequacy for learning tones, which is the goal of this work. Without an explicit framework that encodes the geometry of tonal representations, modelling more complex tonal alternations, such as those discussed above, is more difficult. Establishing this representational foundation is therefore a useful precondition for future work on deeper tonal processes. **[On a second reading, I think this paragraph is still technically correct, but it now feels somewhat outdated. It reads like that we dont know whether transformations can be modeled over ARs, but we now know its possible, and exactly how do to]**

Lastly, we are not aware of existing learning models that operate with ARs directly for either tonal processes or tonotactics. In the next section, we review previous learning approaches related to tier-based representations and learning, and then introduce BUFIA, the algorithm that our model is based on.

2.2. Learning Models

This section reviews systems for phonotactic and tonotactic learning. One line of research employs statistical methods from machine learning (Coleman and Pierrehumbert, 1997; Hayes and Wilson, 2008; Goldsmith and Riggle, 2012). These methods have been deployed on string and syllabic representations for learning segmental phonotactic patterns as well as stress patterns, but have not been used for learning tonotactics with ARs.

Gouskova and Gallagher (2020) synthesises the Maximum Entropy (MaxEnt) strategy of Hayes and Wilson (2008) with a procedure that induces phonological tiers in order to learn non-local phonotactics. Related work by Jardine and Heinz (2016), Jardine and McMullin (2017), De Santo and Aksënova (2021), and Lambert (2021), and Swanson (2025) provide theoretical results and algorithms regarding the induction of phonological tiers for non-local phonotactics. Other work has proposed procedures for learning tiers for grammars modelling non-local phonological processes (Burness and McMullin, 2019, 2021; Belth, 2024). While the successful induction of tiers from data is an important achievement of the 21st century, the constraints induced over the projected tiers are still strings, and not ARs. For example, none of the aforementioned methods can learn a constraint that describes a many-to-one association from the tonal tier to the TBU tier or vice versa.

BUFIA represents another approach (Chandlee et al., 2019; Rawski, 2021). BUFIA differs from the aforementioned approaches in a couple of ways. First, it is not specific to any particular representation or data structure, such as strings or trees. For this reason, Chandlee et al.'s (2019) presentation of BUFIA abstracts away from a specific representation, and presents BUFIA in terms of arbitrary model-theoretic relational structures. As long as the representations of interest can be expressed model-theoretically, BUFIA can be implemented. Since strings, trees, and graphs are all examples of structures that have model-theoretic representations, versions of BUFIA can in principle be designed for finding constraints over any of these representations. This generality not only allows it to be instantiated and ultimately deployed for the learning of a variety of constraint types, but it also makes clear how the structure of the space of possible constraints facilitates learning.

Second, BUFIA is a deterministic learning algorithm and in its current form uses no statistical methods or randomness in its decision-making. Basically, BUFIA checks possible constraints against the data one at a time, adding those that are consistent with the data. It returns a grammar consisting of inviolable constraints. As such, this grammar is a conjunction of negative literals: $\neg C_1 \wedge \neg C_2 \wedge \ldots \wedge \neg C_n$, where each C_i is a constraint, that is, a forbidden relational structure.

As will be explained in Section 3.4, BUFIA works because any representational scheme orders the expressible constraints in a natural way. By beginning with the smallest constraint, BUFIA is guaranteed to search as much of the space as possible and return the most general constraints (that is, the smallest forbidden structures) consistent with the data, before a stopping condition is met. Furthermore, BUFIA has been shown to be at least as successful in learning feature-based, segmental phonotactics of Quechua (Swanson et al., 2026) as phonotactic learning models based on the principle of Maximum Entropy (Hayes and Wilson, 2008; Wilson and Gallagher, 2018). However, BUFIA has not previously been used in tonotactic learning over non-string representations.

To sum up, while ARs have long been conceptualised as graph structures, no existing work learns grammars with ARs, nor has BUFIA been instantiated to operate with ARs. To address this gap, we present BUFIA-AR, a BUFIA-based learner capable of learning tonotactic constraints directly over ARs. To our knowledge, BUFIA-AR is the first learning algorithm that directly infers such constraints from ARs. As an instantiation of the original BUFIA framework, BUFIA-AR inherits its strengths and limitations.

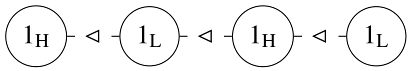
A more detailed presentation of BUFIA-AR is given in Section 4. Before turning to that discussion, we first introduce in Section 3 several elements from formal language theory, model theory, and graph theory that are essential for understanding the algorithm and its internal structure. Some of this background knowledge is well established in computer science (Cormen et al., 2022; West et al., 2001); our presentation is adapted to suit those only familiar with phonological theory.

[I added the paragraph of math disclaimer back but Im just worried that this is what R3 asked us to remove...]

We note that Section 3 and 4 are somewhat technical, and that researchers only interested in BUFIA-AR's performance on the Hausa dataset may find them cumbersome. However, we believe these sections are important to include because they provide the necessary details for someone (1) to understand how BUFIA-AR is an instantiation of the more general BUFIA algorithm, (2) to implement the algorithm themselves (in order to replicate the study we conducted), and (3) to understand how the algorithm's behaviour as well as how it reflects the geometry of tonal phonology.

3. Preliminaries

In this section, we begin with structures for representing strings (Section 3.1), and then extend them to ARs using graphs and matrices (Section 3.2). Central to BUFIA is Model Theory (Enderton, 2001), which views structures as mathematical objects, and uses formal logic to represent structural units and the relationships between them. Model-theoretic analysis has been used to compare differences between different representational theories of phonological structure (Strother-Garcia, 2019; Oakden, 2020;



$$\begin{aligned} \mathcal{M} &= \{D \mid H, L, \lhd\} \\ D &= \{1, 2, 3, 4\} \\ H &= \{1, 3\} \\ L &= \{2, 4\} \\ \lhd &= \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 3, 4\rangle\} \end{aligned}$$

Figure 2: String model for *HLHL* over $\Sigma = \{H, L\}$.

Jardine et al., 2021). The matrix-based representation used here is a common method in mathematics and a particularly convenient way to represent AR graphs. These formal representations are essential for subsequent algorithm development, thereby understanding of the properties and learnability of ARs.

3.1. Strings Representations

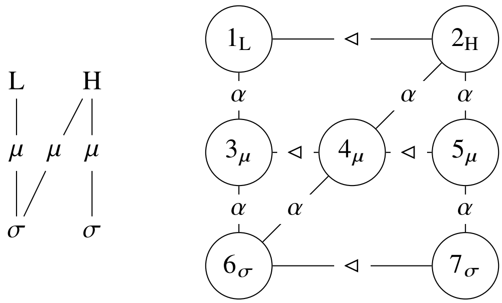
Tonal melodies can be represented as strings, that is, sequences of symbols. Let Σ be a finite alphabet of symbols and Σ^* the set of all strings over Σ. For example, in a language with a tonal inventory $\{H, L\}$, we have $\Sigma = \{H, L\}$, and Σ^* contains all possible strings over Σ, such as *HL*, *HHL*, *HLHL*, and so on. Let $|w|$ denote the length of a string w. For $0 \leq n < m \leq |w|$, let $w[n : m]$ denote a non-empty substring of w starting at position n and ending just before position m. The empty string ϵ is trivially a substring of every string.

A string can also be represented *model-theoretically* with two components: the domain D and a finite set of n-ary relations $\mathcal{R} = \{R_1, R_2, \ldots, R_i\}$ (Rogers et al., 2013; Chandlee et al., 2019; Rogers and Lambert, 2019). A string model of *HLHL* is shown in Figure 2. In this model, $D = \{1, 2, 3, 4\}$ indicates that there are four elements in the string. Positions $\{1, 3\}$ are labelled with the unary relation H, while $\{2, 4\}$ are the positions labelled with L. Elements are connected via a binary *successor* relation, denoted by $\lhd$. For example, $\lhd = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 3, 4\rangle\}$ specifies that position 1 (labelled H) immediately precedes position 2 (labelled L), position 2 immediately precedes position 3, and so on.

3.2. Autosegmental Representation as Graphs

Unlike the string model, where elements are ordered linearly along a single axis, an AR distributes linguistic information across multiple tiers. In this work, in addition to the basic tone and syllable tiers, we incorporate a mora tier, and treat the mora as a secondary unit subordinated to syllables to indicate the syllable weight. Therefore, we assume each syllable must associate with at least one mora. Figure 3a illustrates a disyllabic form with an LH.H melody, which is visualised as a graph in Figure 3b, which corresponds to the model-theoretic relational structure in Figure 3c.

A graph consists of a set of vertices V and edges E. As can be seen in Figure 3c, the graph consists of seven *labelled nodes* which are circles numbered with $\{1, 2, \ldots, 7\}$, representing linguistic units, each labelled with its corresponding properties, such as L, H, σ, or μ. The lines connecting elements are referred to as *edges*, which



$$D = \{1,2,3,4,5,6,7\}$$
$$L = \{1\}$$
$$H = \{2\}$$
$$\mu = \{3,4,5\}$$
$$\sigma = \{6,7\}$$
$$\lhd = \{\langle 1,2\rangle, \langle 3,4\rangle, \langle 4,5\rangle, \langle 6,7\rangle\}$$
$$\alpha = \left\{\begin{matrix}(1,3),(2,4),(2,5),\\(3,6),(4,6),(5,7)\end{matrix}\right\}$$

(a) AR (b) Graph (c) Model

Figure 3: Three representations of a disyllabic LH.H

may be either *directed* or *undirected*. For clarity, we distinguish two types of connections: *association relations*, denoted by α, refer to the undirected edges across tiers; and successor relations, denoted by $\lhd$, represent ordered relationships within a single tier. All the successor connections in the current example are given by $\lhd = \{\langle 1,2\rangle,\ \langle 3,4\rangle,\ \langle 4,5\rangle,\ \langle 6,7\rangle\}$. The association relations, on the other hand, are expressed as a set of unordered pairs: $\alpha = \{(1,3),\ (2,4),\ (2,5),\ (3,6),\ (4,6),\ (5,7)\}$. Together, the tuple $\langle H, L, \mu, \sigma, \lhd, \alpha\rangle$ constitutes the *signature* of this model-theoretic representation of an AR.

In addition to representing model-theoretic relational structures with graphs, one can use *adjacency matrices* (West et al., 2001; Cormen et al., 2022), which use binary values to denote whether there is an edge between two nodes. Two nodes, v and w, are considered *adjacent* if there is an edge connecting them. If G is a graph with n nodes then its adjacency matrix A is an $n \times n$ matrix. If i and j are connected, the entry (i, j) has the value of 1; otherwise 0. Using this method, an AR with t tones, m moras, and s syllables can be encoded as a $t \times m$ tone-mora matrix and a $m \times s$ mora-syllable matrix, with successor relations implicit in the indexing.

For instance, Table 2 provides alternative representation of the same AR in Figure 3. In the tone-mora matrix, the header row reflects the tonal sequence LH, and the index column reflects the mora sequence. Since the L is associated with the first mora μ_1, and the H is associated with both μ_2 and μ_3, the corresponding cells (L, μ_1), (H, μ_2), (H, μ_3) are marked with 1, and others with 0 which indicates no connection between them.[3] If we examine the matrix by columns, one can interpret that the L tone occupies the first mora, while the H tone occupies the remaining two. Likewise, in the mora syllable matrix, since σ_1 is associated with μ_1 and μ_2, and σ_2 with μ_3, the corresponding cells are marked with 1. If we examine this matrix by rows, it can be seen that σ_1 is heavy while σ_2 is light. Hereafter, we refer to the rows and columns in the tone-mora matrix as the mora rows and tone columns respectively; and to the

[3] For convenience, we refer to each cell by the labels of the nodes. Node indices will be omitted from here on. For example, we may refer to $(1,3)$ and $(2,4)$ with (L, μ_1) and (H, μ_2), respectively. However, these are the labels of the nodes rather than the nodes themselves.

rows and columns in the mora-syllable matrix as the syllable rows and mora column respectively.

	L	H
μ_1	1	0
μ_2	0	1
μ_3	0	1

	μ_1	μ_2	μ_3
σ_1	1	1	0
σ_2	0	0	1

Table 2: Adjacency matrices: tone–mora matrix (left) and mora–syllable matrix (right)

These AR adjacency matrices capture precisely the same information as the graphs and model-theoretic structures. If we replace the header row and index column with the node numbers used in the graph model signature, and replace the 1s with the indices of row and column, we obtain the table filled as in Table 3. As can be seen, these pairs are identical to the information encoded in Figure 2. **[It might be more consistent/clear to replace the -> with ◁, as in the one on the left]**

	1	$\triangleleft$	2
3	(1, 3)		0
$\triangle$			
4	0		(2, 4)
$\triangle$			
5	0		(2, 5)

	3	$\rightarrow$	4	$\rightarrow$	5
6	(3, 6)		(4, 6)		
$\downarrow$					
7					(5, 7)

Table 3: Equivalent node-indexed encodings of the adjacency matrices in Table 2, where $\triangleleft$ and α are the successor and association predicates of the model signature.

Furthermore, certain patterns can be easily identified by inspecting the matrices. For example, a series of contiguous 1s may indicate tone spreading over multiple moras if observed in the tone column, or a mora shared by multiple tones if seen in the mora row. Two 1s positioned diagonally suggest a one-to-one mapping. The associations (L, μ_1) and (H, μ_2) in the tone-mora matrix in Table 2 present an example of this. Some particular or ill-formed patterns can also be readily detected, which will be discussed in Section 4.

3.3. Structure Containment

An important concept when considering model-theoretic relational structures, either strings, sequences of feature bundles, or ARs, is *containment*. Intuitively, containment refers to the situation when one structure includes another structure inside of it. We formalise this notion using two related definitions, following Chandlee et al. (2019): *restriction* (Definition 3.1) and *containment* (Definition 3.2). Readers are referred to Rogers and Lambert (2019) for an alternative formulation. In the definitions below, it is assumed the structures A and B have the same signature $\langle R_1, \dots R_n \rangle$.

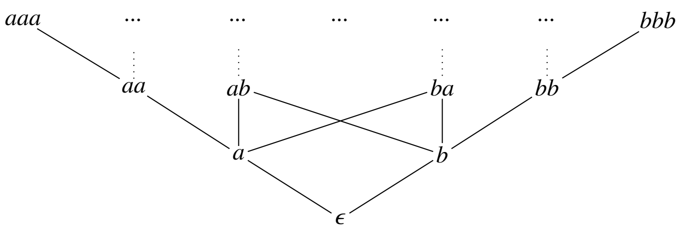


Figure 4: Strings over $\Sigma = \{a, b\}$ ordered by the containment relation (non-exhaustive)

Definition 3.1. *$A = \langle D^A; R_1^A, \ldots, R_n^A \rangle$ is a* restriction *of $B = \langle D^B; R_1^B, \ldots, R_n^B \rangle$ iff $D^A \subseteq D^B$ and for each m-ary relation R_i^A, we have $R_i^A = \{(x_1, \ldots, x_m) \in R_i^B \mid x_1, \ldots, x_m \in D^A\}$.*

Definition 3.2. *Structure A is* contained *in structure B ($A \sqsubseteq B$) if there exists a restriction of B, denoted B', and there exists a function $h : A \to B'$ such that for each m-ary relation R_i in the model signature: $R_i(a_1, \ldots, a_m) \in A \Longrightarrow R(h(a_1), \ldots, h(a_m)) \in B'$.*

These definitions mean that A is *contained* in B whenever there exists a subpart of B onto which A projects via a function h, and furthermore, that the relations that hold among elements of A hold among their corresponding elements in B. For instance, the AR in Figure (**??**) is contained in the AR in (**??**) because the dotted region in (**??**) preserves all internal relations of (**??**), despite the differences in the indicing. In particular, μ_1, μ_2, σ_1 in (**??**) project to μ_2, μ_3, σ_2 in (**??**), respectively.

The containment relation is particularly important because it provides a natural *ordering* over the space of possible structures. The ordering is not total, like a linear order; it is a *partial* ordering. In such a partially-ordered space, some structures are comparable in terms of containment, but not necessarily all of them are. Chandlee et al. (2019) provides some additional terminology: if one structure A is contained in another structure B, then structure A is a *subfactor* of structure B, whereas B is a *superfactor* of A. They prefer these terms to substructures and superstructures, which have different technical meanings in model theory.

To illustrate, Figure 4 displays a non-exhaustive portion of the partial ordering of possible strings over the alphabet $\Sigma = \{a, b\}$. At the bottom of the ordering is the empty string ϵ, with more complex strings appearing higher in the ordering. Whenever two elements are connected by a line, it means the higher element contains the lower element. For example, no line relates the a to b, because neither contains the other. However, a line is drawn from a to aa, ab and ba, because a is contained in each of these structures. As defined above, aa, ab, ba and aaa, and any other strings that contain a are superfactors of a; correspondingly, a is a subfactor of each of these strings.

There can be many layers between a superfactor and one of its subfactors. When there are no structures ordered between a superfactor and its subfactor, it is called a *next superfactor*. In this sense, among all superfactors of a such as $\{aa, ab, ba, aaa, aab, \cdots\}$, only those superfactors that are exactly one level above are next superfactors of a, namely $\{aa, ab, ba\}$.

3.4. BUFIA

Recall that given a set of positive examples, BUFIA returns a grammar just like the ones considered by Jardine (2017): a conjunction of negative literals: $\neg C_1 \wedge \neg C_2 \wedge \ldots \wedge \neg C_n$. BUFIA begins its search of the possible structure space with the most general structure (the empty structure) and progresses to more specific ones. In other words, it proceeds from the bottom of the inverted pyramid upwards. In this process, two functions are central to BUFIA: Contain and NextSupFact. For each candidate structure S, BUFIA applies the Contain function to check whether it is attested in the positive data. If it is, BUFIA concludes it is not a bona-fide constraint, discards it, and uses NextSupFact to generate a set of next superfactors to continue the search. If, on the other hand, S is absent from the positive data, it is added to the grammar as a constraint. For structures that are labelled as constraints, BUFIA does not apply NextSupFact. Consequently, all structures/possible constraints that contain it are pruned from the remaining search. The learning process continues until a stopping condition is met, which in this work is a predetermined upper bound on the size of the constraints.

Using the same example from Figure 4, suppose the language forbids adjacent bs, formalised as a constraint $^*bb$, which excludes strings $\{bb, bba, bbb, abb, \cdots\}$. For concreteness, assume a data set D equal to $\{aba, baabaa\}$. BUFIA begins from the most general hypothesis: applies Contain to check whether the empty string ϵ is attested in the positive data. Since ϵ is contained in every string trivially, Contain returns true, and BUFIA does not add ϵ as a constraint. Then, BUFIA applies NextSupFact to generate the next superfactors of ϵ, which are the two singletons a and b. In a breadth-first order, BUFIA checks whether a and b are contained in the positive data. In the same fashion, Contain function confirms both strings are attested as substrings of D, discards them, and then generates their superfactors for later checking. When BUFIA reaches bb, Contain returns false because bb is not contained in any datum in the data set (neither aba nor $baabaa$ contains bb). BUFIA therefore adds $^*bb$ to the grammar and skips generating any superfactors of bb. The search then continues with the remaining candidates until the stopping condition is met (e.g., the size of structures exceeds a predefined size such as two).[4]

Chandlee et al. (2019) prove that BUFIA returns a set of constraints that (1) are consistent with the data (no constraint returned by BUFIA is contained in any datum in the positive data); (2) contain the most general constraints (obtains the simplest possible constraints according to the order imposed by the containment relation, e.g. only $^*bb$ is returned but not $^*bba$, $^*bbb$ or others) and (3) generates the smallest set of structures consistent with the data, among the other possible grammars that are consistent with the data subject to the same conditions (generalises conservatively).

One concern for BUFIA is its time complexity: BUFIA runs in exponential time in the worst case (Chandlee et al., 2019). However, as Swanson et al. (2026) point out, data sparsity is highly advantageous for learning since the hypothesis space is thus pruned more efficiently.

[4] An alternative stopping condition is to set a number of constraints. Both kinds of stopping conditions control the amount of generalisation. The longer the search is permitted, more constraints may be found, and amount of generalisation is reduced. Without any limits, it will overfit the data. However, when the search terminates sooner, generally fewer constraints will be found, and the grammar makes broader generalisations.

Another concern raised by the reviewers is that, in its present form, BUFIA is fragile with respect to noise. For example, if there is a single data point violating a constraint C in the data (a "noisy intrusion"), then C will not be included in the output grammar, even if C is an otherwise valid generalisation. It is important to put this current limitation in context. First, even in the absence of noise, it is currently unknown how to learn constraints expressed with ARs. From the viewpoint of scientific practice, it makes sense to address an easier problem before addressing a harder problem. Second, BUFIA can be adapted to handle noise in different kinds of ways. Prior literature suggests multiple strategies for handling noisy intrusions, exceptions and sparse data (Angluin and Laird, 1988; Yang, 2016; Wu and Heinz, 2023; Kaelbling et al., 2023; Dai, 2025). Considering versions of BUFIA that are robust to all kinds of noise is a valuable direction for future research. Third, it is worth pointing out that the mere use of statistical methods for learning does not automatically make learners robust to noise (Angluin, 1988). In other words, noisy data is an issue, and one that can be studied separately from the fundamental issue of successful generalisation. See Heinz (2009) for additional discussion on factoring learning problems.

4. Learning over Autosegmental Representations

This section introduces the BUFIA-AR in Section 4.1. Sections 4.2 and 4.3, address its two core functions, NextSupFact and Contain, are realised specifically for ARs. Those sections will also demonstrate that the use of adjacency matrices simplifies the computational implementation, making it more straightforward and interpretable. Our goal is to walk through each piece step by step, as they are essential for (i) understanding the logical reasoning behind tonotactic learning over ARs, (ii) properly interpreting the results that follow, and (iii) replicating our study or adapting BUFIA-AR for other AR-like structures in future work.

4.1. BUFIA-AR

BUFIA-AR instantiates BUFIA with ARs, and the Contain and NextSupFact functions operate in the same manner as in Section 3.4 but only differ in representations. In the case of BUFIA-AR, the stopping condition is defined as predetermined upper bounds on the numbers of tones, moras, and syllables. These steps are formalised in Algorithm 1, and are discussed in more detail below.

BUFIA-AR initially requires three inputs: a finite set of positive data over ARs D, an initial empty AR ar_0, and a set of maximum constraint limits $t, m, s \in \mathbb{N}$. The positive data D contains a finite set of well-formed ARs. The positive integers t, m, s limit the numbers of tones, moras, and syllables of an AR considered by the algorithm. Additionally, the algorithm initialises two empty sets: G, representing the grammar which is empty at first and will be returned with constraints discovered by the algorithm once the learning is complete; and a set V which stores checked ARs to prevent the learner from revisiting superfactors of those that have already been examined.

A queue Q is used to store structures awaiting to be checked by the algorithm, following a standard breath-first search method. As shown in Line 5 and 11, there are two

Algorithm 1 BOTTOM-UP FACTORIAL INFERENCE ALGORITHM OVER AUTOSEGMENTAL REPRESENTATIONS

Require: Positive data D, initial empty AR ar_0, maximum tier limits t, m, s
Ensure: Set of learned generalisations G

$Q \leftarrow ar_0$
$G \leftarrow \emptyset$
$V \leftarrow \emptyset$
while $Q \neq \emptyset$ **do**
  $ar \leftarrow Q.\textsc{dequeue}()$
  if $ar \notin V$ **then**
    $V \leftarrow V \cup \{ar\}$
    if $\exists x \in D$ s.t. $ar \sqsubseteq x$ **then**
      $\mathcal{A} \leftarrow \textsc{NextSupFact}(ar)$
      $\mathcal{A}' \leftarrow \{ar' \in \mathcal{A} \mid \text{size}(ar') \leq (t, m, s) \wedge \nexists g \in G \text{ s.t. } g \sqsubseteq ar'$
      $Q.\textsc{enqueue}(\mathcal{A}')$
    else
      $G \leftarrow G \cup \{ar\}$
    end if
  end if
end while
return G

operations related to Q: DEQUEUE and ENQUEUE: the former takes an AR structure ar out of the queue for checking against the data, and the latter adds the next structure into Q, making it available for the learner to evaluate later in a breath-first manner.

BUFIA-AR is called with the initial structure ar_0 set to the most generalised and least complex structure, the empty structure. The CONTAIN function takes the current structure as an argument and checks whether it is a subfactor of the data. If so, the NEXTSUPFACT function generates its next superfactors which must also satisfy the following conditions: the tone, mora, and syllable of AR do not exceed t, m, s, have not yet been visited, and do not contain any structures already forbidden by the grammar. The eligible superfactors will be "enqueued" for later checking. If the current structure is determined to be missing from the dataset—that is, it is not a subfactor of any data point—then it is added to G. This process repeats until the size of the structure exceeds the predetermined threshold t, m, s.

As mentioned, function CONTAIN and NEXTSUPFACT are crucial in the implementation of BUFIA-AR. In the following sections, we will first introduce how to expand a given AR to get its next superfactors (Section 4.2). Then we explain how to check whether one AR is contained within another (Section 4.3).

4.2. How to Expand Autosegmental Representations

Here we explain how we compute the next superfactors of an AR. This can be achieved by adding new units to tiers or by adding associations between tiers. In a general graph-theoretic setting, any addition at any position would in principle be allowed when constructing a supergraph. However, for ARs, two principles must be taken into account: linguistic meaningfulness and computational efficiency.

By linguistic meaningfulness, we require that the resulting graphs be linguistically interpretable and obey basic universal constraints on phonological representations. For example, expansions that result in inconsistent tier projections, such as adding a mora to the tonal melody tier or line crossing, will be excluded. Some linguistically invalid expansion examples are shown in Figure 5 with dotted lines indicating impossible associations.

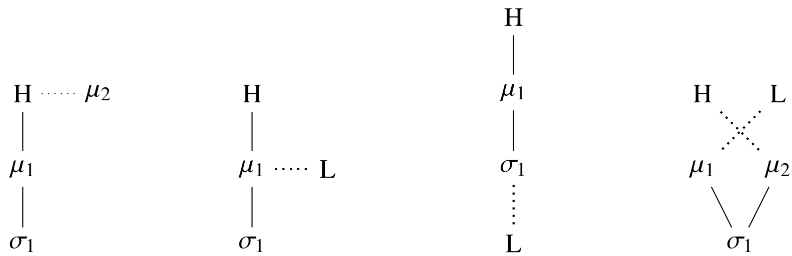


Figure 5: Invalid expansions

By computational efficiency, we aim to obtain distinct graphs while minimising redundant construction of identical structures. In many cases, the same resulting AR can be obtained via multiple expansion paths. For example, inserting a mora between two existing moras may yield the same structure as appending a mora to the right edge of the tier or prefixing one to the left edge. Such redundant derivations incur unnecessary computational cost by repeating equivalent work, and we therefore seek to avoid them.

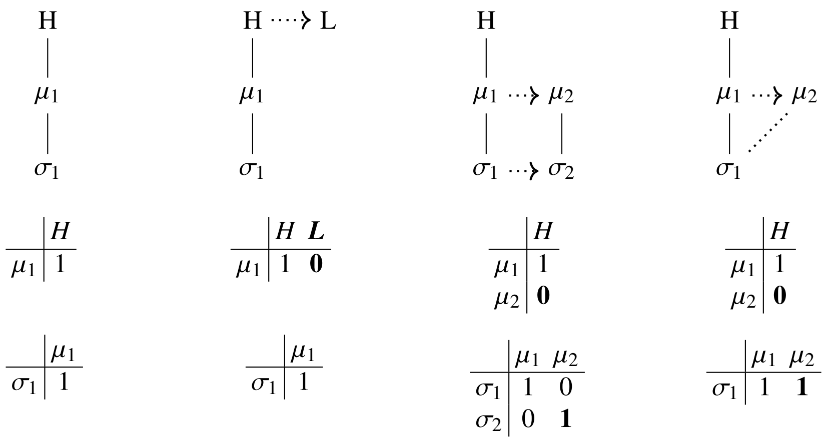
4.2.1. Adding Units

Based on these considerations, we require that new elements be added only to their projected tiers and only append them to the right edge, consistent with the temporal relation. This yields three specific Expansion Rules (ERs):

- **ER1** Append a tone to the tone tier.
- **ER2** Append a mora to the mora tier.
- **ER3** Append a monomoraic syllable to the syllable tier.

ERs (1–3) specify how a new tone, mora, or syllable can be appended to the end of each tier, analogously to extending a string. Figure 6 illustrates how the base AR in Figure 6a can be expanded in three different ways: by adding a tone L (6b); by adding a monomoraic syllable σ_2, which by default associates to a single mora μ_2 (6c); or by adding a mora to the rightmost syllable (6d).

One might wonder how ER1 relates to the Obligatory Contour Principle (OCP). There is cross-linguistic evidence for pressures to avoid sequences of identical tones, although whether the OCP should be a universal rule has been debated (Goldsmith,



(a) Base AR (b) Add a tone (c) Add a syllable (d) Add a mora

Figure 6: Base AR and its superfactors by adding a new tone, syllable, and mora

1976; Meyers, 1997). Our case study in Hausa enforces the OCP by only appending a tone different from the last tone in the current AR. There are two reasons for this practice. First, there is no clear evidence that argues against the OCP in Hausa; second, enforcing the OCP also helps reducing the search space by restricting tonal sequences. Nonetheless, relaxing this restriction is possible and is left for future research.

4.2.2. Adding Association Lines

Besides adding units, an AR can also be changed by modifying an association line and leaving the number of the segments or features unchanged. This ability to manipulate associations independently of segments forms the core mechanism of autosegmental theory. Concerning the association lines, Goldsmith (1976) proposes a set of well-formedness conditions (WFCs) as a guidance of how association lines may be added or deleted. These conditions require association changes to be minimal while maximising satisfaction of basic representational requirements.

1. All vowels are associated with at least one tone.
2. All tones are associated with at least one vowel.
3. Association lines do not cross (the No-Crossing Constraint, NCC).

From a computational perspective, unrestricted growth of association lines is costly, as it allows arbitrary connections between any two nodes. The NCC plays a crucial role in reducing this complexity by limiting the set of admissible association lines. There have been several works providing formal definitions of the WFCs (Goldsmith, 1976; Coleman and Local, 1991; Jardine and Heinz, 2015; Jardine, 2017). Based on these, the NCC rule can be formalised as following. Letting α be the set of existing associations, a new association (i, j) is valid if and only if there does not exist an association $(i', j') \in \alpha$ such that $(i' > i \wedge j' < j) \vee (i' < i \wedge j' > j)$.

At the same time, to further avoid adding association lines which give rise to identical graphs, we impose an additional constraint for computational efficiency: an association line may be added only to the the first unassociated unit. In other words, a fully-specified AR (all tones, moras, syllables are already associated with at least one element), will not add new association lines. If an AR contains more than one unassociated unit, the new association must be formed to the first available unassociated unit. This is to satisfy WFC 1 and 2. Finally, from a linguistic perspective, we assume that the association lines between moras and syllables are pre-specified. Consequently, new association lines are formed only between moras and tones. This yields the ER4:

ER4 Add an association line between tone and mora, subject to the following conditions:

a. do not violate the No-Crossing Constraint;
b. involve at least one unassociated unit; and
c. be formed adjacent to the final mora–tone association.

The final tone-mora association in an AR can be expressed mathematically as in Definition 4.1.

Definition 4.1. *An edge* (t, m) *in the tone–mora matrix is defined as the* final *tone–mora association if* $t \in H \cup L$, $m \in \mu$, *and* $(t, m) \in \alpha$, *and there exists no edge* $(t', m') \in \alpha$ *such that* $t' > t$ *and* $m' > m$.

Once the final tone-mora association is identified at (t, m), there are three options to form an association line (t', m') while adhering to ER4:

Definition 4.2. *Given the final tone–mora association* (t, m), *a new edge* (t', m') *adjacent to* (t, m) *can be formed in three ways:*

a. $(t', m') = (t{+}1,\ m)$ *if* $t{+}1 \in D$
b. $(t', m') = (t,\ m{+}1)$ *if* $m{+}1 \in D$
c. $(t', m') = (t{+}1,\ m{+}1)$ *if* $t{+}1,\ m{+}1 \in D$

For example, the AR in Figure 7a contains an unassociated H tone and two unassociated moras, μ_3 and μ_4, as reflected by zeros in the tone–mora matrix. Taking the final association (L, μ_2) as the reference point, three new ARs can be formed by adding one adjacent association: (i) linking the unassociated tone to the nearest mora (Figure 7b); (ii) linking an unassociated mora to the nearest tone (Figure 7c); or (iii) directly linking an unassociated tone and an unassociated mora (Figure 7d).

These operations offer several advantages both computationally and linguistically. Computationally, we control the growth of the next superfactors (by reducing the number of times a unique structure is produced) without sacrificing expressiveness. We do not claim the expansion rules are optimal, only that they outperform naive approaches to superfactor generation. Linguistically, these operations generate a full tonal inventory, including light, heavy, super-heavy syllables, as well as simple and contour tones and tone-spreading patterns.

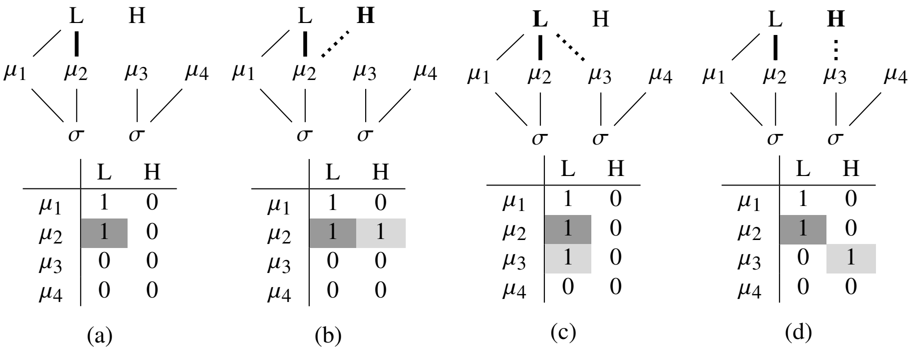


Figure 7: Three options to form new associations

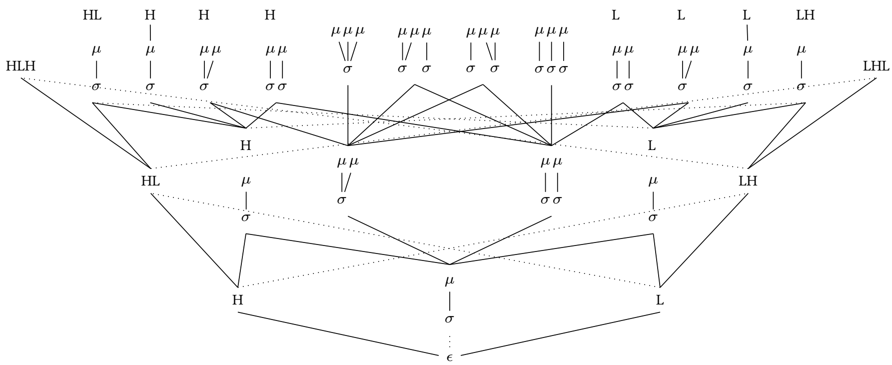


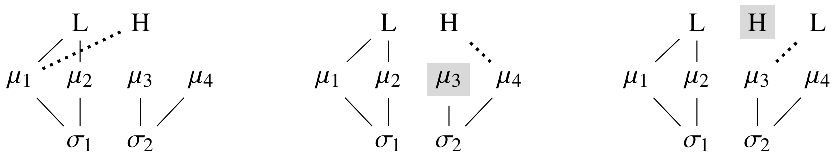
Figure 8: A partially-ordered hypothesis space over ARs

Recall from Section 3 that factors stand in a containment relation, whereby superfactors contain subfactors. This relation yields a hypothesis space, illustrated in Figure **??**, in which lines (some shown as dotted for readability) connect a subfactor (lower-ranked) to its next superfactors (higher-ranked). If two factors are not connected by any path in this graph, they are incomparable, such as the unassociated H and the unassociated L.

4.2.3. Interim Summary

To summarise, this subsection has presented a concrete method for computing the next superfactors of a given AR. Four Expansion Rules were defined: they expand the structure either by adding a unit on a single tier or by introducing a new association.

We used adjacency matrices to represent the internal associations, and stipulated some rules to reduce redundancy. Specific matrix patterns were shown to diagnose ill-formed structures. For example, Figure 9 illustrates three invalid ARs. An anti-diagonal (from upper right to lower left) of 1s marks a line-crossing violation (9a). A column or row that is filled exclusively with zero indicates an unassociated tone or mora. Such zero-domination is problematic when it occurs between two columns or rows that contain associations, as this pattern indicates that an intervening unit has been skipped, resulting in an improper association formed across it (as in (9b) and (9c)) as they violate the WFC 1 and 2.



	L	H
μ_1	1	1
μ_2	1	0
μ_3	0	0
μ_4	0	0

(a) lines crossing

	L	H
μ_1	1	0
μ_2	1	0
μ_3	0	0
μ_4	0	1

(b) mora skipping

	L	H	L
μ_1	1	0	0
μ_2	1	0	0
μ_3	0	0	1
μ_4	0	0	0

(c) tone skipping

Figure 9: Invalid association lines

An alternative interpretation of skipped units is that they are *floating*. We distinguish between *floating* and *unassociated* units. Floating tones, which are common in African languages, are not associated to any TBU and manifest themselves under specific phonological conditions. Unassociated units, by contrast, are available to be linked during the expansion process.

BUFIA-AR does not express floating tones *per se*, only unassociated units. In principle, floating tones could be incorporated by either introducing them into structures explicitly, or by interpreting unassociated units as floating. We also leave this matter for future research.

It should be emphasised that explicitly defining how ARs are expanded is crucial for two reasons. First, the expansion procedure forms the basis of BUFIA-AR: without it, the space of possible constraints cannot be traversed, or even represented, and thus the learnability of AR constraints cannot be properly understood. Second, this procedure highlights the multilinear structure and associations between autosegmental tiers, which are hallmarks of tonal phonology.

4.3. Containment Relation

Section 4.2 introduces the four ways to obtain next superfactors of an AR. This section addresses a different question: given two ARs, how can we determine if one AR *contains* the other?

This question is important for any implementation of BUFIA because it provides an explicit, formal criterion for checking containment rather than an impressionistic intuition. For example, one might assume that a light HL contour is contained in a heavy HL syllable, and similarly that a light LH is contained in a heavy LH. Under this assumption, any language that allows a heavy contour would necessarily also allow the corresponding heavy contour. This is, however, not accurate. As shown in Table 4, Hausa accepts heavy HL but not light HL, whereas Mende accepts heavy LH but not light LH.

In other words, the algorithm must be able to determine that none of the structures in Table 4 contains one another. Containment between ARs is not determined solely by tonal string matching or by the number of TBUs, which are straightforward to verify (e.g., L is contained in HLH, and a bimoraic sequence is contained within a trimoraic sequence). A more challenging task is to identify if the associations across tiers (i.e., the temporal arrangement of autosegments) in one AR are fully preserved in another.

	light LH	**light HL**	**heavy LH**	**heavy HL**
AR	L H \ / μ | σ	H L \ / μ | σ	L H | | μ μ \ / σ	H L | | μ μ \ / σ
Hausa	N	N	N	Y
Mende	N	Y	Y	Y

Table 4: Contour Inventory in Hausa and Mende

There is more than one method to achieve this algorithmically. We use a matrix-based containment checker to determine whether two ARs, even with the same tone sequence and the same number of syllable and mora, will have the same associations.

The matrix containment follows exactly parallel to the general notion of any structural containment: matrix A is *contained* in matrix B if there exists a *submatrix* of B, denoted B', that functions as a restriction discussed above, such that all relations within A are relatively preserved in correspondence with those in B'. Definitions 4.3 and 4.4 are adaptations of Definitions 3.1 and 3.2.

Definition 4.3. *Let matrix A be an* $m \times n$ *matrix. Matrix B is a* submatrix *of A by taking* k $(1 \leq k \leq m)$ *contiguous rows and* v $(1 \leq v \leq n)$ *contiguous columns from A.*[5]

Definition 4.4. *An* $m \times n$ *matrix* $A = [a_{ij}]$ *is* contained *in a matrix* $B (A \sqsubseteq B)$*, if there exists an* $m \times n$ *submatrix in B, denoted by* $B' = [b'_{ij}]$ *such that for all* $i \in [0, m], j \in [0, n]$, $b'_{ij} \geq a_{ij}$.

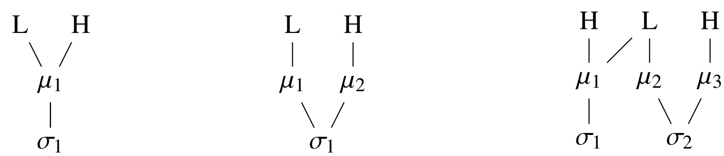
Definition 4.3 defines what a submatrix is, which is analogous to the general restriction definition in Definition 3.1. Definition 4.4 states that a smaller matrix $A = [a_{ij}]$ is contained in a larger matrix $B = [b_{ij}]$ if we can find a submatrix $B' \subseteq B$, with the

[5]The term "submatrix" is generally defined by deleting arbitrary rows and/or columns from the original matrix. However, in our case, we require a more restricted definition: the remaining rows and columns must be *contiguous* in the original matrix, which reflects tonal locality.

same dimensions as A, such that the value of each entry in B' is at least as large as the corresponding entry in A; that is, $b'_{ij} \geq a_{ij}$ for all i, j. Since the entries of these matrices are binary, this condition implies two cases:

(a) If $a_{ij} = 1$, meaning there is an association between node i and node j in the smaller matrix A, then b'_{ij} must also be 1. This ensures an association found in A is preserved in B' at the same position;
(b) If $a_{ij} = 0$, meaning no association exists between node i and node j in the smaller matrix A, then b'_{ij} may be either 0 or 1. In other words, B' may or may not have an association at the corresponding position. It is permitted to contain additional associations beyond those in A, but it must not omit any that are present in A.

The comparison proceeds by first identifying whether the tonal string of one structure is contained within another, and then verifying whether the corresponding association matrices align. Even when a tonal substring match is found, containment is established only if the associated tone–mora and mora–syllable submatrices match within the aligned region. The full algorithm is provided in the Appendix, along with a detailed walk-through. Figure 10 illustrates this procedure: (a) is not contained in (b) or (c), whereas (b) is contained in (c) since the submatrix in (c) (the shaded cells) matches matrices of (b), in terms of associations as well as tonal strings, but does not match those of (a).



(a)

	L	H
μ_1	1	1

	μ_1
σ_1	1

(b)

	L	H
μ_1	1	0
μ_2	0	1

	μ_1	μ_2
σ_1	1	1

(c)

	H	L	H
μ_1	1	1	0
μ_2	0	1	0
μ_3	0	0	1

	μ_1	μ_2	μ_3
σ_1	1	0	0
σ_2	0	1	1

(a) Light rising (b) Heavy rising (c) A superfactor of (b)

Figure 10: Adjacency-matrix encodings of a light rising tone (a), a heavy rising tone (b), and a superfactor of the heavy rising tone (c).

4.4. Summary

This section established BUFIA-AR, which is grounded in autosegmental theory. In particular, we explained the reasoning and implementation of two functions that are core to the BUFIA-AR framework: NextSupFact and Contain, which are used in the case study described in the next section.

5. Case Study: Hausa

This section presents a case study applying BUFIA-AR to identify tonotactic constraints over ARs in Hausa. We examine Hausa because suitable data was available; we are unaware of a similar dataset for Mende. The analysis focuses on non-derived native Hausa words, examples of which are shown in Table 5. As mentioned earlier, in Hausa monomoraic syllables can bear either H or L, but cannot bear contours. A bimoraic syllable can take an HL contour (each tone associates to one mora) but LH contours are not allowed. Lastly, Hausa does not have superheavy syllables, therefore monosyllabic three-tone contours are also not allowed.

[the special letters won't show]

Monosyllabic	áa 'son'	mèe 'what'	mâi 'oil'	
Disyllabic	á.sáa 'land'	mù.tûm 'person'	tú.dùu 'hill'	màa.gée 'cat'
Trisyllabic	í.nú.wàa 'shadow'	àn. á.ráa 'ice/snow'	múu.jì.yáa 'owl'	fà.dá.màa 'swamp'

Table 5: Nonderived native Hausa words (Awagana et al., 2009)

Newman (2002, p.606) notes non-derived disyllabic words exhibit canonical normal tone patterns: H.L, L.H, and H.H. For trisyllabic nouns, H.L.H, H.H.L, L.H.L, and L.H.H are very common, but L.L.H is less common with basic lexical items. Words ending in L.L, either disyllabic or polysyllabic words containing L.L, are relatively uncommon, especially when the second syllable bears a long vowel. When L.L forms do occur, they are typically restricted to ideophones or loanwords (Newman, 2002, p.606). Zoll (2003) argues that the absence of H.L.L and L.L.H is a result of the constraint *Lapse (which forbids L from spreading across adjacent syllables) outranking *Clash (which forbids H from spreading across two syllables). In addition, for canonical, non-derived forms falling (F) tones typically fall on the last syllable (Newman, 2002, p.606).

Based on the above, the linguistically attested constraints in Hausa are summarised in Table 6, which are translated into AR constraints in Table 7. In this study, we treat these constraints as the baseline grammar against which performance of BUFIA models is evaluated. Specifically, we investigate whether BUFIA-AR can accurately learn AR-based constraints, and if so, how the computationally attested constraints (henceforth, BUFIA constraints) correspond to previously identified linguistic ones (baseline constraints). We also include a BUFIA-string model as a comparison, which learns constraints over the same data using string representations.

Generalisations	Interpretation
*RISE$_\sigma$	Syllables may not be associated with rising tones (LH)
*CONTOUR$_\mu$	Moras may be associated with at most one tone.
*3T-CONTOUR	No trimoraic contour
*NONFINAL-FALL	No nonfinal HL contour for polysyllabic words
*LAPSE	No L spreading across syllables

Table 6: Phonological Generalisations in Hausa

*RISE$_\sigma$	*CONTOUR$_\mu$	*3T-CONTOUR	*NONFINAL-FALL	*LAPSE

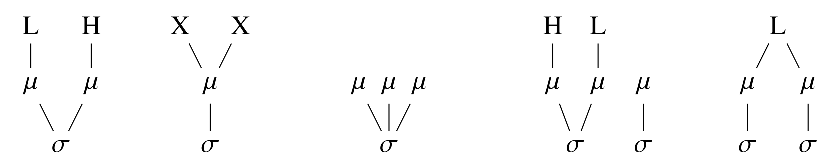


Table 7: Phonological Generalisations Translated into forbidden ARs

5.1. Experiment Setup

The data used for the current experiment comes from the Hausa dataset (Awagana et al., 2009) in The World Loanword Database (WOLD) (Haspelmath and Tadmor, 2009). It consists of around 1460 core words whose meanings are present in the Loanword Typology meaning list. This list includes forms and concepts that all languages have terms for. For the Hausa dataset, the vocabulary contains 1668 core meaning-word pairs, along with other information in fields titled Form, Meaning, Loanword Status, and Analysability. To ensure consistency with previous generalisations, which are based on canonical, non-derived tonal patterns, we restrict our dataset to monomorphemic words that are clearly non-borrowed, and all others were filtered out. In addition, forms containing glides (w, y, kw, ky, gw, gy) **[special letters won't show]** were excluded to avoid ambiguous syllabification and uncertain syllable weight. As a result, a total of 621 words transcribed in orthography remained and were stored in a .csv file.

All orthographic forms were first syllabified (Kodner, 2016) then converted into its corresponding AR representation using the method described in Jardine and Heinz (2015). Table 5.1 shows how an orthographic form with tonal markers is transformed into its corresponding matrix representation. After conversion and removal of duplicate ARs, 64 distinct AR types were identified among the 621 words, which serve as the positive dataset on which BUFIA-AR operates.

[I want to remove the matrix-coding from the table]

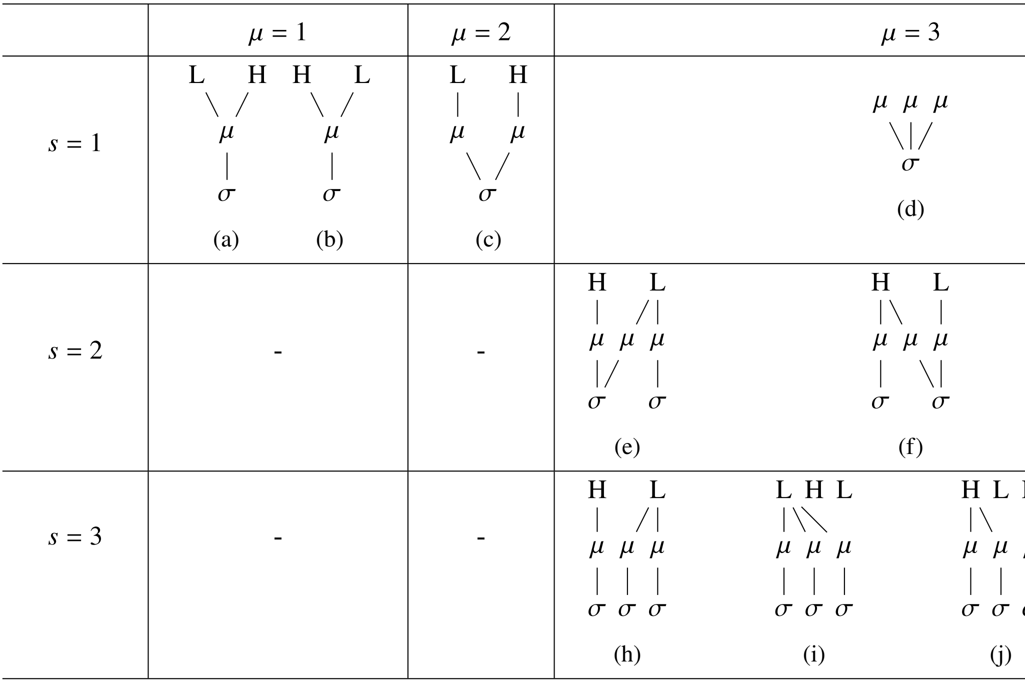


Table 8: Hausa constraints returned by BUFIA-AR when $s \leq 3,\ t \leq 3,\ \mu \leq 3$

the melody and syllabification implicated by the structures in Figure 8. Boldface indicates the tone on heavy syllables. These labels serve only as shorthand for the ARs shown in Table 9 and may be ambiguous, as they do not fully capture the structures.

Constraints Present in both BUFIA and Baseline	**Baseline Constraints Absent from BUFIA**	**BUFIA Constrai Absent from Base**
*Rise$_\sigma$ (8c)	*NonFinal-Fall	*F.L (8e)
*Contour$_\mu$ (8a–8b)	*Lapse	*H.F (8f)
*3T-Contour (8d)		***H**.H-LH (8g)
		*H.L.L (8h)
		*L.L.L-HL (8i)
		*H.H-LH (8j)
		*L.H.H-L (8k)

Table 9: Comparing baseline constraints with BUFIA constraints

The first category is when the BUFIA-AR attested constraints align with the linguistic generalisations previously argued for in the literature. For monosyllabic forms (when $s = 1$), BUFIA-AR returned four constraints: a monomoraic LH (8a), a monomoraic HL (8b), a bimoraic LH (8c), and a trimoraic structure (8d). Each of these matches a baseline constraint. In particular, (8a) and (8b) together account for the absence of monomoraic contour tones, or the constraint $*$Contour$_\mu$. The two constraints further rule out any structures in which a single mora associates with more than two tones. Secondly, the BUFIA-constraint (8c) precisely forbids a heavy LH contour, which is identical to $*$Rise constraint in Table 7. Its counterpart, a heavy HL syllable, is not returned, which appropriately addresses the contour asymmetry problem discussed earlier. Finally, (8d) correctly disallows trimoraic syllables as expressed by the constraint $*$3T-Contour. BUFIA-AR does not return any specific tonal forms for this structure, since any tonal sequence it carries will ultimately violate the forbidden structure.

One thing to note here is that both Contour$_\mu$ and 3T-Contour describe structural information: Contour$_\mu$ prohibits binary branching on a mora node, while 3T-Contour prohibits ternary branching on a syllable node. The reason that 3T-Contour can be ruled out by a single constraint (8d), whereas two constraints, (8a) and (8b), are needed to account for Contour$_\mu$ is that in the current implementation, tonal specifications must be fully specified, and we did not include an umbrella feature, or permit under-specified tonal nodes, that would match both H and L.

Overall, BUFIA constraints (8a-8d) correctly reflect frequently reported patterns regarding the tonal inventory, TBU, and syllable weight, and arguably cover some of the most fundamental aspects of Hausa tonology. This demonstrates that BUFIA-AR is not only capable of learning these tonotactic patterns but also expressing them in terms of ARs.

The second category regards baseline constraints without matching BUFIA constraints. There are two of these constraints, $*$NonFinal-Fall and $*$Lapse. Each of these appear to be partially, not totally, accounted for by some of the BUFIA constraints. For example, constraint (8e) partially captures the generalisation $*$NonFinal-Fall, which requires that HL contours do not occur in non-final position, and in particular, when the following syllable bears a L tone. This is a clue that the BUFIA constraints appear to allow for the possibility of an HL contour followed by a H syllable because no constraint banning such a configuration was returned. Indeed, a reverse search revealed six words that exhibit these structures, as shown in Example (1).[6] These data points indicate that the constraint $*$NonFinal-Fall is too strong and it must be further refined if it is to account for the full range of observed data.

(1) Hausa words exhibiting HL.H pattern ($*$NonFinal-Fall)
  a. jîn. ái “the pity”
  b. kûn.née “the ear”
  c. tsûm.máa “the handkerchief or rag”
  d. sâi.wáa “the root”
  e. bâu.táa “worship”
  f. yân.yáa.wàa “the fox (fennec of Sahara)”

[6] Newman (2002, p. 607) lists eight non-derived words with an initial HL tone, possibly reflecting the historical loss of a low-tone vowel. The word *kûn.née* ‘the ear’ is one of these; the others did not appear in Newman’s records.

Consider the next baseline constraint ∗Lapse, which disfavours the L-tone spreading, therefore predicts the absence of H.L.L and L.L.H (Zoll, 2003). The BUFIA constraint (8h) *H.L.L bans one form of H.L.L when all three syllables are monomoraic, but does not entail banning other H.L.L structures, such as ones that contain a heavy syllable. An example in the data is the word *sá.bòo.dà* 'because.' Even though this word expresses the H.L.L pattern, it does not contain the structure shown in (8h) and therefore it satisfies BUFIA constraint (8h) *H.L.L.[7]

Furthermore, BUFIA-AR did not return a constraint prohibiting the factor L.L.H. In fact, there are also instances of this pattern as shown in Example (2).

(2) Hausa words exhibiting the L.L.H pattern

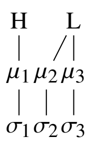
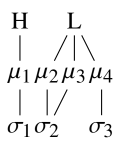
a.	mà.là.fáa	"the hat or cap''
b.	tùn.kù.yáu	"the flea''
c.	gì.zàa.góo	"the adze''
d.	tàu.ràa.róo	"the star''
e.	gàa.jì.màa.rée	"the cloud''

To summarise, the reason BUFIA did not find the baseline constraints ∗NonFinalFall and ∗Lapse is simply because they are not true of the data, which has multiple forms violating those constraints.

This brings us to the third category, which includes BUFIA constraints without matching baseline constraints. Generally, these constraints partially match ∗NonFinalFall and ∗Lapse. However, since both baseline constraints are not generally true of the data, the BUFIA constraints are typically more specific refinements of them, forbidding narrower classes of forms. We can classify these constraints to two types: constraints concerning contour (8e, 8f) and constraints concerning spreading (8g, 8i, 8j, 8k).

Constraint (8e) and (8f) specify two environments that forbid a HL contour: when a HL syllable is followed by a light L-toned syllable (8e, *F.L), or when it is preceded by a light H-toned syllable (8f, *H.F). The forms of F.L and H.F, termed as "non-contrasting" tone by Zoll (2003), have been reported in other languages. For instance, in Mende, disyllabic R.H undergoes tone absorption and simplifies to L.H, and F.L simplifies into H.L (Leben, 2017). In Hausa, since the light syllables with rising tones are banned, BUFIA-AR only returns *F.L and *H.F constraints. Newman (2002, p.598) points out that F.L optionally changes to H.L in complex relative pronouns and adverbs; however, such simplification does not apply to our dataset, with which BUFIA-AR remains consistent.

In terms of the constraints concerning spreading, traditional OT includes ∗Clash and ∗Lapse, which ban adjacent syllables linked to either H or L. Among BUFIA constraints, ∗Clash and ∗Lapse are reflected in more specific forms as in constraint (8g-8k).

[7] Visually,  might appear to be contained in  but in fact the two structures are incomparable in the partially-ordered space because μ_3 is associated to σ_3 in the former but not in the latter.

Constraints (8e, *F.L), (8h, *H.L.L), and (8i, *L.L.L-HL) address the cases which forbid L spreading in particular circumstances. Constraint (8e) and (8h) prevent it after H tones, and (8i) prevents a trisyllabic L span before a HL melody.

Similarly, the constraint (8f, *H.F), (8g, ***H**.H-LH), (8j, *H.H-LH), and (8k, *L.H.H-L) restrict H-spreading in different environments. For example, (8g) and (8j) prohibit an H tone (preceding an LH) from spreading across two syllables, while (8k) forbids an H tone from spreading to two syllables when it is wedged between two L-toned syllables.

Finally, four BUFIA constraints (8g,***H**.H-LH), (8i,*L.L.L-HL), (8j,*H.H-LH), and (8k,*L.H.H-L) potentially demonstrate the expressive advantage of ARs over string representations. These constraints involve unassociated units, either unassociated tones or syllables. For example, Constraint (8g) describes the following condition: when a H spreads over a heavy syllable to another syllable, the tonal tier can only be extended by adding one more tone. The association of the unassociated tones does not have to be specified. A similar pattern is seen in (8i) where if L tone spreads over three light syllables, the tonal tier will only accommodate one more tone. These patterns reflect a central fact regarding autosegmental phonology: units from different tiers do not need to be synchronised. This idea is more difficult to express with purely linear string representations.

In summary, this subsection reports the qualitative evaluation of BUFIA-AR by comparing BUFIA constraints with the constraints reported by linguists. It was found that BUFIA-AR is able to learn tonotactic patterns over AR, and that there are interesting differences between the constraints produced by BUFIA-AR and those posited by earlier researchers.

5.2.2. Quantitative Evaluation: Cross-validation

The k-fold cross-validation procedure partitions the dataset into k folds. In each iteration, BUFIA-AR is trained on $k-1$ folds and evaluated on the remaining held-out fold. This process is repeated k times so that each fold serves as the evaluation set exactly once.

Since BUFIA is a deterministic categorical learner, the distinction between token- and type-based datasets does not ultimately affect the learned grammar. However, we report both settings for completeness. In the type-based setting, the dataset consists of 64 distinct Hausa AR types, randomly divided into four folds (16 types per fold), with 48 used for training and 16 for testing. In the token-based setting, the full dataset of 621 tokens is also randomly divided into four folds (155/156 per fold). Within the train set (465/466 in total), there were 55–62 unique types.

In each round, BUFIA-AR is trained on three folds and evaluated on the remaining fold. The baseline grammar is evaluated on the same held-out fold. Accuracy is defined as the proportion of correctly accepted forms in the held-out set (also known as *recall*).

In addition, we apply the same procedure using a string-based version of BUFIA (BUFIA-Str). Both BUFIA-AR and BUFIA-Str require a user-defined complexity bound, which determines the maximum size of constraints considered by the algorithm. However, this notion of complexity is not directly comparable across representations. For BUFIA-AR, it is controlled by multiple parameters (t, m, s) that limit different

	Type-Based				Token-Based			
Fold	**Train/Test**	**AR**	**Str**	**Baseline**	**Train/Test**	**AR**	**Str**	**Baseline**
1	48/16	0.938	0.500	0.812	465(55)/156	0.987	0.949	0.936
2	48/16	0.875	0.500	0.812	466(60)/155	0.994	0.974	0.942
3	48/16	0.875	0.812	0.938	466(62)/155	1.000	1.000	0.955
4	48/16	0.938	0.812	0.688	466(55)/155	0.994	0.942	0.974
Avg		**0.907**	**0.656**	**0.812**		**0.994**	**0.966**	**0.952**

Table 10: Four-fold cross-validation in both type- and token-based settings. In the token setting, the number of distinct types in Train are indicated in parentheses.

tiers; we set these to $t = m = s = 3$ to match the constraints found in 8. BUFIA-Str uses a single complexity parameter k. Since the maximum constraint length in Table 9 is 7 (i.e. 8j), we set $k = 7$ to match this bound. This setting allows us to assess whether constraints identified over AR can also be recovered using a string-based representation under a comparable complexity limit.

As shown in Table 10, BUFIA-AR outperforms BUFIA-Str and the baseline grammar, and reaches an average accuracy of 0.907 in the type-based setting and 0.994 in the token-based setting. The overall performance increase in the token-based setting is expected because of the uneven distribution of tokens over types: most types are rare (Yang, 2013). Consequently, frequently attested structures are likely to be attested in training and form a larger proportion of the testing data.

BUFIA-Str performs below the baseline in the type-based setting (0.656 < 0.812), but outperforms in the token-based setting (0.966 > 0.952). The lower performance in the type-based setting is likely due to the relatively high complexity threshold ($k = 7$), which leads to overfitting in the type-based setting. In the token-based setting, this effect is compensated. These results suggest that the BUFIA-AR supports stronger generalisation over distinct types, including rare ones, even without frequency information.

In addition, across all four folds, BUFIA-AR consistently returns the constraints (8a–8f) and (8h), while the remaining constraints vary slightly across folds. Altogether, this indicates that BUFIA-AR not only captures the core generalisations of Hausa tonotactics but also extends to rarer and more exceptional patterns that the baseline fails to account for. Overall, the quantitative evaluation demonstrates that BUFIA-AR generalises robustly to unseen data and empirically outperforms a similar algorithm using string representations and a theoretically motivated baseline grammar.

6. Discussion

6.1. Optimisation versus Satisfaction

The grammars output by BUFIA-AR compute the well-formedness of a word by checking whether its structure *contains* any forbidden structures (the negative literals), or equivalently, whether it *satisfies* all of the constraints. This is in contrast to globally *optimising* over *violable* constraints. The pattern obtained by satisfying the 11 BUFIA constraints captures the asymmetric distribution of spreading and contour formation while accommodating the rarer data points. On the other hand, In Zoll (2003)'s Optimal Tone Mapping program, tonal mapping arises from the optimisation over morphological directionality constraints (ALIGN-R and ALIGN-L) and quality-sensitive markedness constraints (∗CLASH and ∗LAPSE).

What are we to make of these differences? Consider how each approach explains the common absence of non-contrasting forms such as ∗F.L and ∗H.F. Optimal Tone Mapping utilises ∗CLASH and ∗LAPSE to help explain the absence of these unattested forms. Zoll observes this is an improvement over the traditional mapping approach, which explained their absence by the lack of a motivation for spreading. Zoll (2003) writes, "with one-to-one linking built into the Association Convention, nothing automatically forces one tone to spread onto a syllable that already bears tone." On the other hand, BUFIA-AR returns no ∗LAPSE or ∗CLASH constraints, but instead returns more specific versions of them. There appears to be a trade-off between optimisation-based grammars using simpler constraints and satisfaction-based grammars using more complex constraints. One may then prefer optimisation-based grammars over satisfaction-based ones on the grounds that its constraints are simpler.

However, we argue that such a conclusion is premature. First, from a computational perspective, this trade-off may not be as significant as it appears because the phonological patterns being described are ultimately finite-state (Karttunen, 1998) or subregular (Heinz, 2011a,b). There are many ways to decompose a phonological pattern into parts, and it has been argued that understanding each of these ways is valuable (Lambert, 2025).

Second, it is also the case that the dataset obtained from Awagana et al. (2009) shows that Zoll's (2003) proposal, as it is presented, cannot adequately capture the exceptional data points in the dataset because *LAPSE is undominated. Zoll acknowledges in footnote 18 that exceptions to the general pattern in Hausa could be accommodated by a prespecificational account (Inkelas and Cho, 1993). However, another alternative that Zoll does not consider would be to add to the grammar refined versions of *LAPSE like the ones that BUFIA-AR found. In other words, even optimisation-based grammars would benefit from more complex constraints of the variety found by BUFIA-AR.

One objection to this latter approach is that it stands in opposition to a core idea behind Optimality Theory, which is that CON is a universal set of simple constraints and all language-specific variation is a product of the interaction of these constraints via ranking. Under this perspective the only learning that needs to take place regards the ranking of the constraints (Tesar and Smolensky, 1998, 2000). This argument is fine as far as it goes, but we are aware of no theory of CON which provides a simplicity criterion its constraints must satisfy.

It is also the case that more recent research has shown how constraints themselves can be learned from examples (Hayes and Wilson, 2008; Heinz, 2009, 2010; Lambert et al., 2021). This line of work coheres with the idea that not all constraints in all languages are universal, and that some are in fact learned. Provided a theory of generative grammar admits the learning of some language-specific constraints, it is not clear to us why it would be a problem for language-specific constraints that are a little more complex than the ones attributed to Con to be included in a grammar.

In this regard, it is important to point out that BUFIA-AR does not return arbitrarily complex constraints. The collection of constraints output by BUFIA-AR forms a grammar which corresponds to a particularly weak form of logic (the conjunction of negative literals), and the structures themselves (ARs) are representations well-motivated by linguistic theory.

In sum, the debate between optimisation-based and satisfaction-based approaches is one with a long history (see for example Simon (1956)), and will not likely be settled soon. Nonetheless, the fact remains that BUFIA-AR returns a set of inviolable constraints consistent with the data comparable to the ones utilised in earlier analyses.

6.2. Gradience versus Categoricity in Grammars

Another ongoing debate in phonotactic learning concerns whether the grammar should be categorical or gradient. Learning models like BUFIA(-AR) make clear distinctions between what is well-formed and what is not since constraints are inviolable. By contrast, gradient learning models, such as MaxEnt (Hayes and Wilson, 2008; Gouskova and Gallagher, 2020) discussed in Section 2.2, treat constraints as *weighted*. In these models, Constraints extracted from a lexicon are assigned with probabilistic weights that correspond to well-formedness judgements from native speakers, with lower probabilities representing lower degrees of acceptability. In this line of research, gradience is inherent to the grammar, and probabilistic induction is seen as inevitable for phonotactic learning (Hayes and Wilson, 2008; Wilson and Gallagher, 2018).

We argue that the absence of probabilistic induction in BUFIA-AR should not be regarded as a deficiency for two main reasons. First, whether weights are *necessary* in learning or in the grammar remains a matter of debate. Second, although the presented implementation of BUFIA-AR is categorical, it can be augmented with probabilistic values if a gradient model is desired in certain learning scenarios.

Regarding the first point, several lines of argument have challenged the view that phonotactic grammars are gradient. Directly equating probability with well-formedness, as Hayes and Wilson (2008) do, has been argued to be mistaken because unlikely outcomes can still be well-formed and likelihood ultimately decreases as size increases, but well-formedness does not (Heinz and Idsardi, 2017). Others have argued that gradience in acceptability judgements may stem from performance-related factors, such as misperception (Kahng and Durvasula, 2023) or psycholinguistic processing (Gorman, 2013). Additionally, there is some evidence that categorical models can perform just as well as, and sometimes even better than, gradient models (Gorman, 2013; Durvasula, 2020; Kostyszyn and Heinz, 2022; Dai, 2025; Swanson et al., 2026; Durvasula, 2025). For example, Swanson et al. (2026) conducted a comparative study between BUFIA and MaxEnt on Quechua (Wilson and Gallagher, 2018) and obtained comparable results. Durvasula (2025) demonstrates that when MaxEnt weights are either equalised or entirely removed, the model can actually yield stronger predictions. Other recent studies further highlight the versatility of categorical learners: Payne (2024) shows how BUFIA can be modified to learn positive grammars, while Dai (2025) proposes a categorical learning model capable of filtering out exceptions.

Another point of view that has been expressed is that the gradient/categorical distinction is not intrinsic to phonotactic learning or the phonotactic grammar itself, but only differs in how grammars assign values to constraints. In other words, the categorical and gradient approaches can be analysed the same way; only the values that a learner or grammar associates with constraint evaluation changes (Goodman, 1999; Chandlee and Heinz, 2017; Mayer, 2025). We are sympathetic to this view because this reconciliation foregrounds grammatical structure and backgrounds the particularities of the values. The values (whether binary, scalar, or real) are details, but the grammatical structure itself is paramount. This is precisely what theoretical computational studies strive to elucidate (Heinz, 2018).

This is why it is worth noting that the current categorical learning model of BUFIA-AR can be flexibly extended into a gradient learning model. Doing so only requires that a different value-assigning function be applied to the constraints. Accordingly, the current BUFIA-AR model can be adapted into a gradient version in multiple ways. Weights could be assigned to constraints obtained by BUFIA-AR on the basis of any number of calculations including frequency of the factors, their expected frequency, or based on a regression fit to these or related numbers. Another approach would be to make the ARs themselves gradient. In the current implementation, the binary values (0/1) in the matrices simply represent the presence or absence of association lines, which treats all types of associations in one of these two ways. These binary values can be replaced with probabilistic values of specific associations, so as to get a gradient grammar (Smolensky et al., 2014; Mayer, 2021).

Finally, we emphasise that, despite the long-standing shift in tonal phonology from segment/feature-based representations to non-linear representations, existing gradient learning models do not, as far as we are aware, operate directly over non-linear representations such as ARs. We believe that it will be easier to obtain a gradient tonotactic learning algorithm over ARs by modifying BUFIA-AR as opposed to adapting existing string-based gradient learning models to ARs.

In sum, the categorical nature of BUFIA-AR does not prohibit it from being an important contribution to our understanding of how tonotactic patterns can be learned.

7. Conclusion

This paper presents the first tonotactic learning algorithm over autosegmental representations to our knowledge. Through a case study of Hausa, we constructed a database containing 64 distinct ARs from 621 surface-true Hausa words and discovered, using BUFIA-AR, a tonotactic constraint grammar consisting of 11 constraints (limited to three syllables, three tones, and three moras). Furthermore, by comparing BUFIA constraints with the baseline constraints, we found partial overlaps: in some cases, the two matched, but more often the BUFIA constraints were more specific. We also found data points in conflict with previous analyses, and identified novel patterns not previously reported in the linguistic literature, such as certain constraints prohibiting tone spreading and contour formation in particular environments. These findings demonstrate both the feasibility of tonotactic grammar inference over ARs and the advantages of using algorithmic approaches to refine and extend earlier generalisations.

Another contribution of this paper is the modelling of ARs, including their expansion and containment checking, in a manner that is both computationally efficient and linguistically faithful. The pipeline employed in this study, from data preparation and orthography-to-AR conversion to the implementation of BUFIA-AR, can serve as a general tool for exploring tonotactic constraints in other languages, and as a foundation for future work on more complex tonal process learning.

There are several directions for future research. It would be valuable to investigate the psychological reality of the constraints inferred by BUFIA-AR, seeking external evidence from cognition and perception studies. Such evidence could bear on the stopping criteria for BUFIA-AR. Another future direction is to learn tonal processes. Yolyan (2025) presents a method within the framework of Boolean Monadic Recursive Schemes (Chandlee and Jardine, 2021) for learning phonological transformations over segments represented with features. Since her approach makes use of BUFIA-style generalisation and model-theoretic relational structures, it becomes an interesting question how her approach can be extended to input-output mappings over ARs.

Finally, although this paper primarily targets tonotactic patterns in African languages, the proposed autosegmental learning framework naturally extends to Asian tonal languages such as Shanghai and Fuzhou Chinese, where tonal distributions are tightly conditioned by syllable structure and segmental features (Yip, 2002). It would be interesting to explore how to model such an enriched tonal geometry, and how tonal grammars could be learned over ARs in these languages.

Competing interests. The authors declare no conflicts of interest regarding the publication of this paper.

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Data availability statement. Replication data and code can be found in GitHub repository: https://github.com/lihan829/BUFIA_AR.git.

Ethical standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author contributions. HL curated data, conducted the case study, and wrote the first draft of the manuscript. HL and JH revised the manuscript and approved the final version.

Supplementary material. State whether any supplementary material intended for publication has been provided with the submission. **[So we have an apeendix - does it count as supplementry material?]**

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